



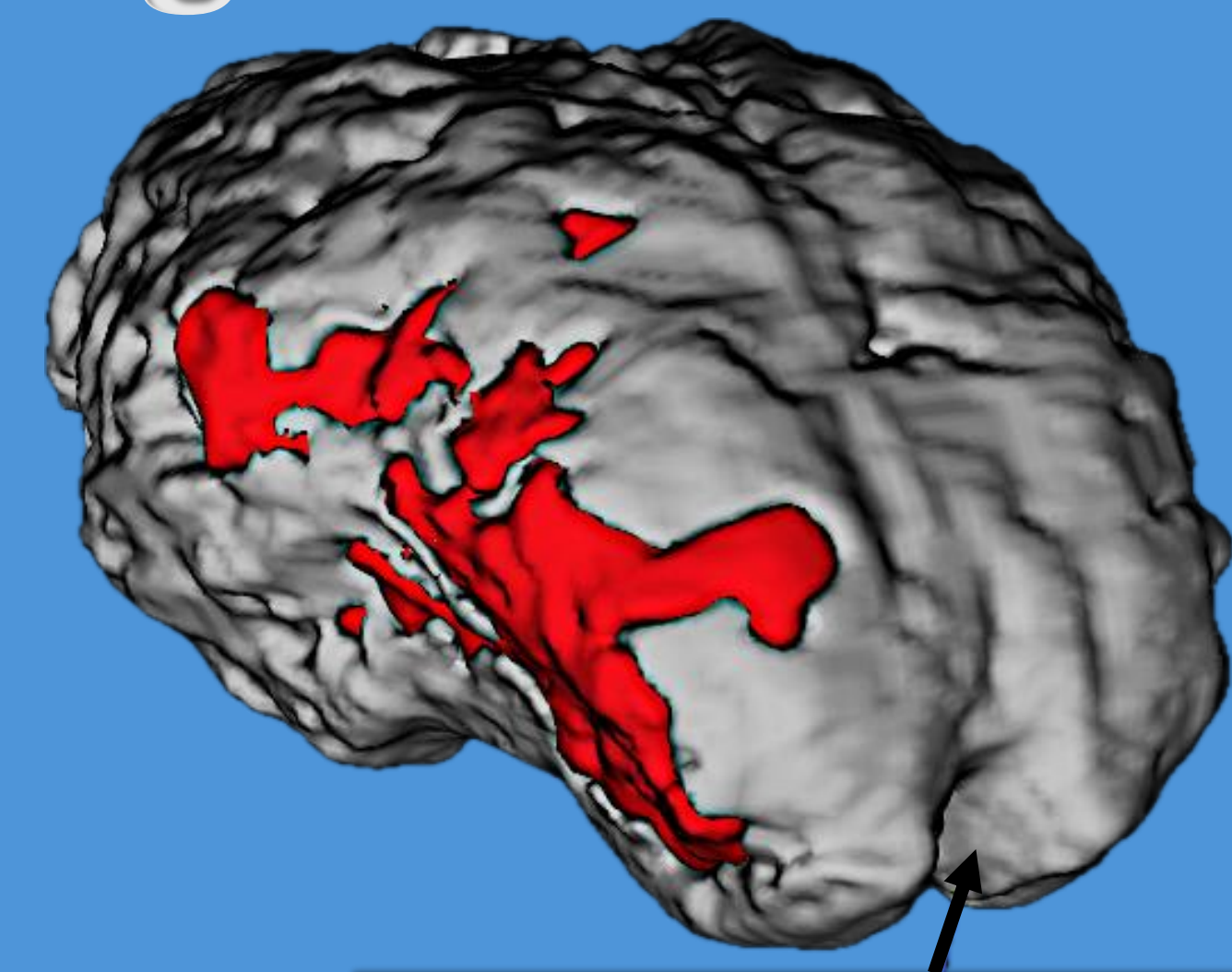
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A Pipeline for the preprocessing of MRI data in an *Ovis aries* stroke model using Python



Biotechnology and
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1. Introduction

Magnetic resonance imaging (MRI) methods allow for structural, functional, and connectivity analysis of the brain in health and disease. Much of this research is conducted on humans and consequently many of the preprocessing algorithms are optimized for human anatomy.

Stroke is the third leading cause of human mortality and morbidity globally (3) and can be studied in animal models for the development of new treatments such as thrombolysis, neuroprotective agents (4), and aiding rehabilitation. MRI is sensitive for the detection of stroke on diffusion weighted imaging (5) modalities and allows for detailed analysis of changes in the brain with T1, T2 and fMRI methods.

Therefore, there exists a need for preprocessing algorithms of non-human MRI data. Much of the current software makes assumptions about the shape of the image that suit humans and give inadequate results for non-human subjects. This poster presents a method for an important preprocessing step for MRI data, namely 'skull-stripping', in which non brain tissue is removed from the image. The 'Brain Extraction Tool' (BET) from the FSL software suite (6) is commonly used for the 'skullstripping' preprocessing step in human data and is compared with the algorithm developed here. The pipeline is automated and can be scaled up for large amounts of data. The code for the pipeline is made available at the bottom of the poster via a QR code to a GitHub repository.

2. Methods

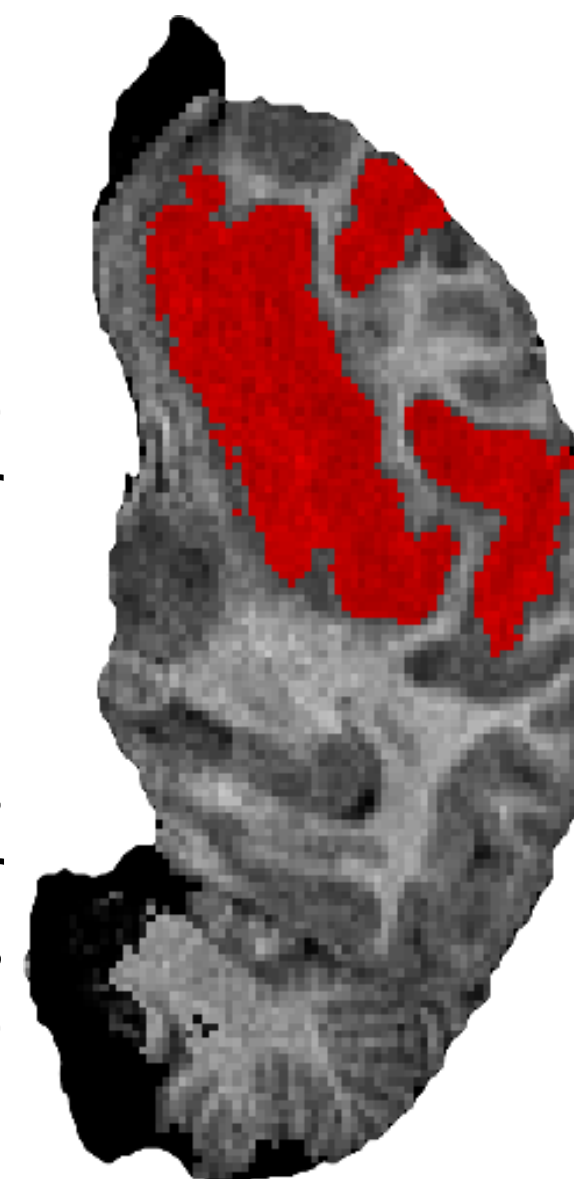
The pipeline is written in python programming language, displayed and annotated in Jupyter notebook (available to view via the QR code). Libraries used include matplotlib for the figures and 'neuroimaging in python pipelines and interfaces' (Nipype)(7) for wrapping tools from different software packages, primarily 'advanced normalization tools' (ANTs) and FMRIB software library (FSL) (see figure 5).

20 sheep have undergone surgically induced stroke by applying a clip to the middle cerebral artery, and MRI images were acquired at pre-surgery, 24-hours, 3-, 6- and 28-day timepoints. Structural T1 images (seen in figures 1-3) and functional MRI (figure 5b) were recorded, alongside other modalities and the use of contrast.

To achieve skullstripping of the raw data, a study-specific template of the 20 pre-surgery images was created using the 'MultivariateTemplateConstruction2.sh' call from ANTs via the terminal. Each scan was then registered using ANTs Registration_Synquick (8) to this template to generate linear transformation and warp files (figure1). The inverse of these transformations were then applied to a binary 'mask' created manually in ITK-snap from the first pre-surgery sheep scan (figure2), using the 'ApplyTransforms' call from ANTs. When this mask has been brought into alignment with the native anatomy it is used to skullstrip the original image via FSL's 'ApplyMask' call. This process is repeated for each scan which has been retrieved by the 'select files' node from Nipype, and the result is stored via a datasink node.



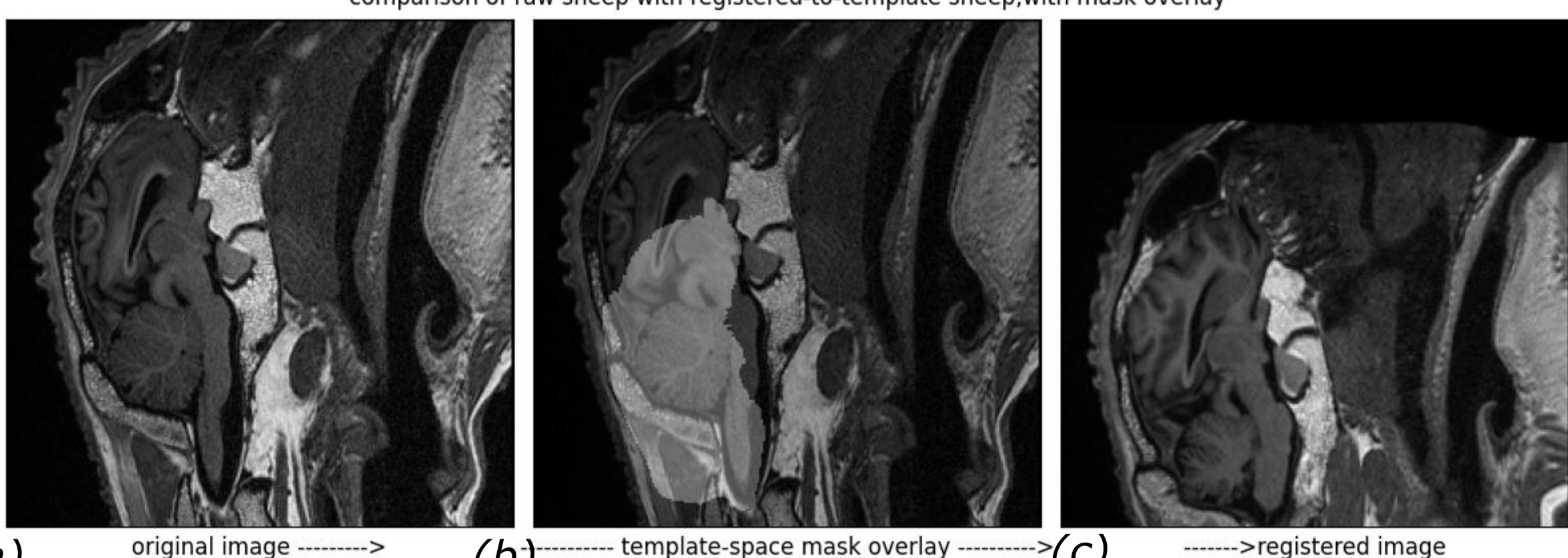
(axial section of RT090 scan with stroke labelled in red)



(sagittal section of RT090 scan with stroke labelled in red made with ITK-snap on data at the 24hour timepoint)

3. Results

On directing the selectfiles node to a directory containing the raw 100 scans (20 sheep at 5 timepoints each) the pipeline will run for several hours. It achieved a tight brain extraction on all 100 scans on manual inspection (figure 3c 'Skullstripped RT090 brain'). In comparison with the FSL BET tool (which is optimized for human data), there is an improved result (figure 4). With the addition of an FSL min step (which selects the smallest dimension fMRI volume in a timeseries) and a dilate-image step, the pipeline can be repurposed for fMRI data, which also requires skullstripping before analysis.



comparison of raw sheep with registered-to-template sheep, with mask overlay

Figure 1 : raw scan image of sheep RT090 (a) is compared with the template created from an average of 20 pre-surgery sheep in light grey (b). The RT090 image is seen aligned with the template in (c). This calculation generates the necessary transformation and warp files that will be inverted and applied to the mask in figure 2.

Check the result of the above transformation aligned with the raw image after the inverse warp and inverse linear transform (aim is to get red mask aligned with background image of brain)

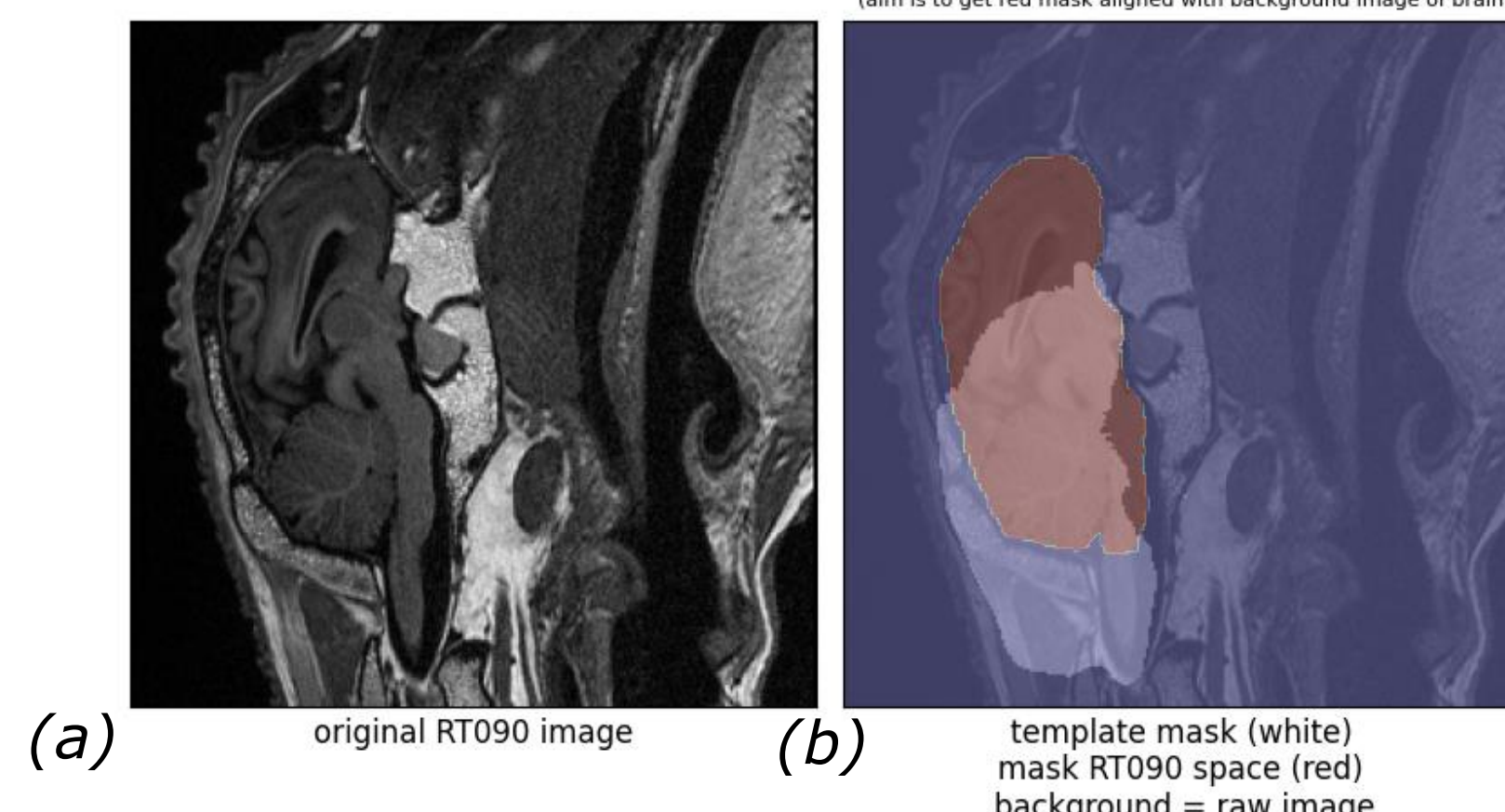


Figure 2 : raw scan image of sheep RT090 (a) the template in light grey (b) and the template after the inverse transformations have been applied in red (b).

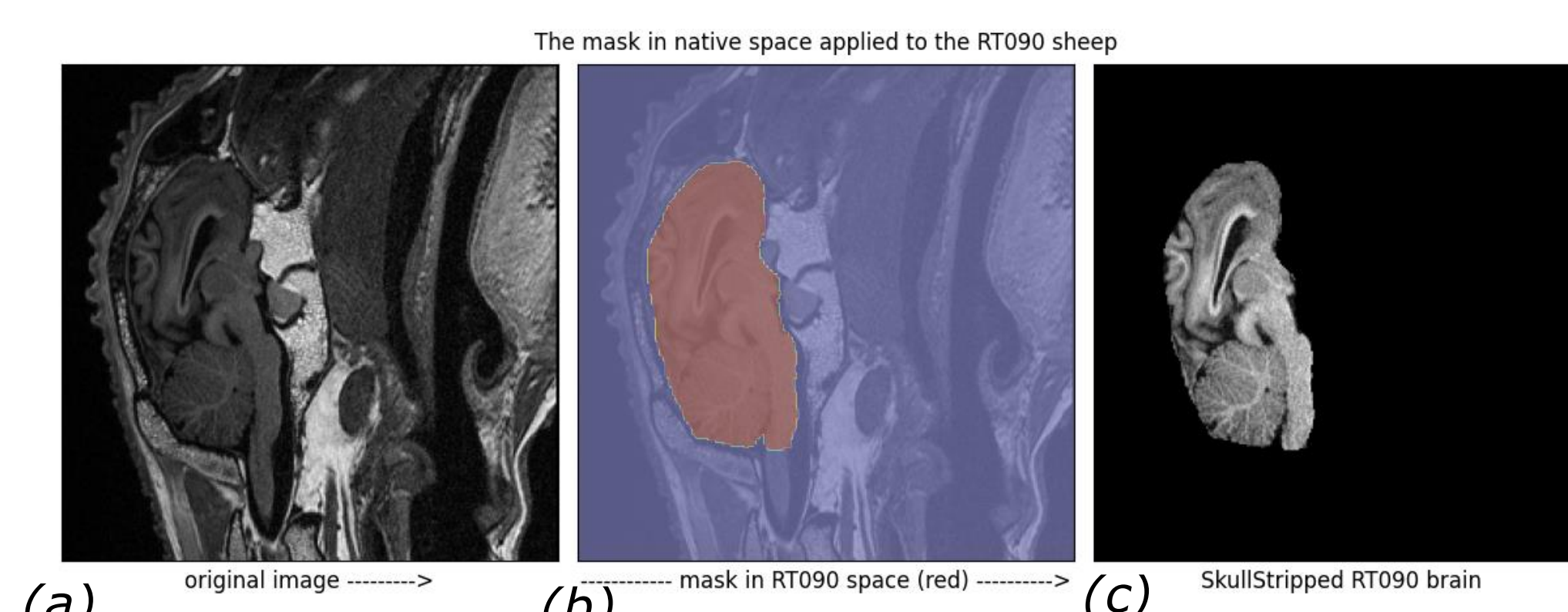


Figure 3 : raw scan image of sheep RT090 (a) aligned template in red overlaid on the original image (b) and the skullstripped result after 'ApplyMask'(c).

4. Conclusion

As diseases such as stroke are an acute emergency there is a need for an animal model. This model requires preprocessing techniques that therefore work on non-human subjects, as demonstrated here. The advantages of this pipeline include a reproducible result in one script that utilizes diverse software packages (6, 8). It is scalable, and while it takes several hours to run on 100 images, there is scope for leveraging cluster or cloud computing to reduce runtimes. It could be quickly repurposed for fMRI images (figure 5b) and the results were run through FSL's FEAT tool for statistical analysis of fMRI data, with adequate result with a sensible output of the anatomical to functional registrations and MCFLIRT (FSL's motion correction tool). Disadvantages include the use of both linear and nonlinear registration to align the mask with each individual sheep, which is computationally expensive; while FSL's BET tool only takes seconds to run per scan. Future directions include use of the skullstripped fMRI data and structural scans for analysis of the functional connectome in stroke at several timepoints, investigating recovery and rewiring of the brain after stroke. It would also be of interest to see if the pipeline works on other species, such as rodents, as these are also commonly used stroke models.

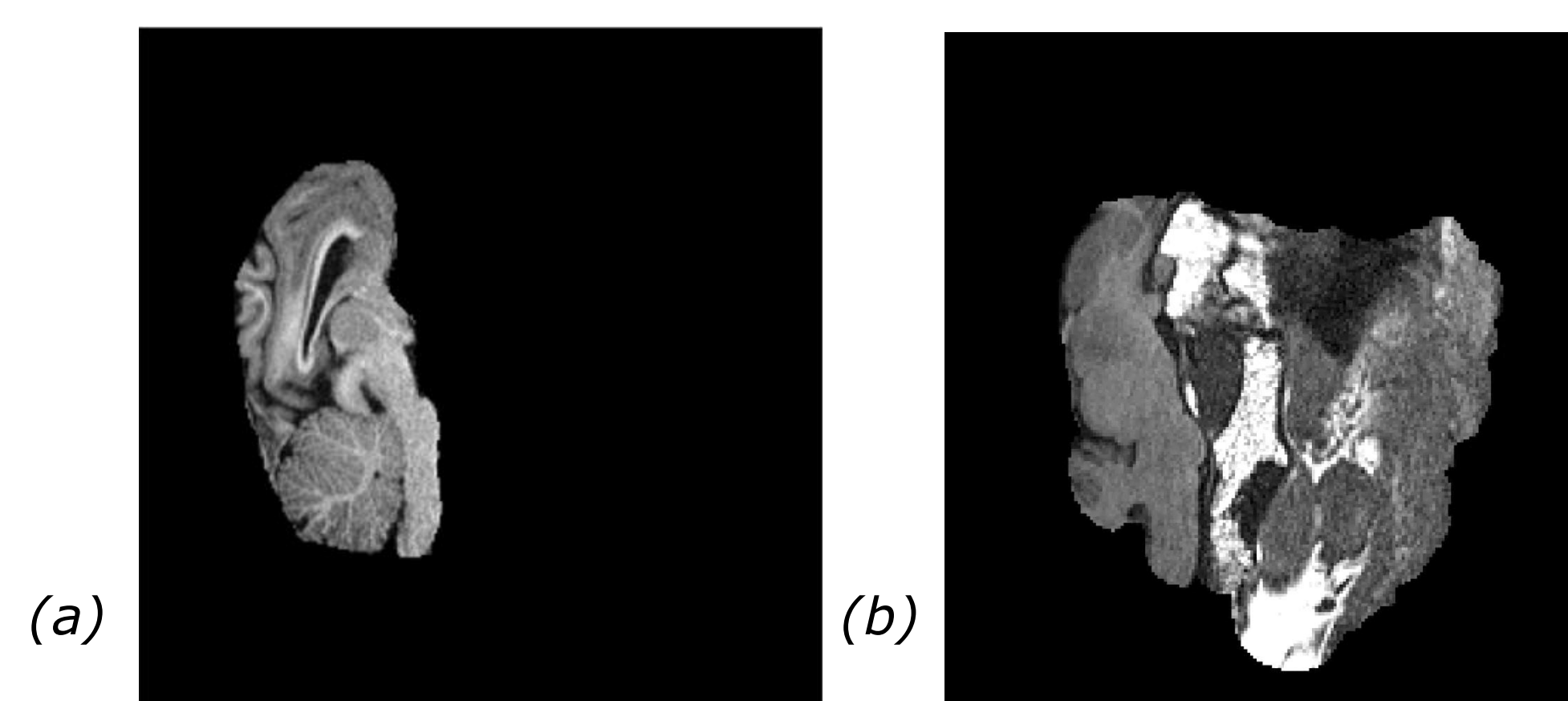
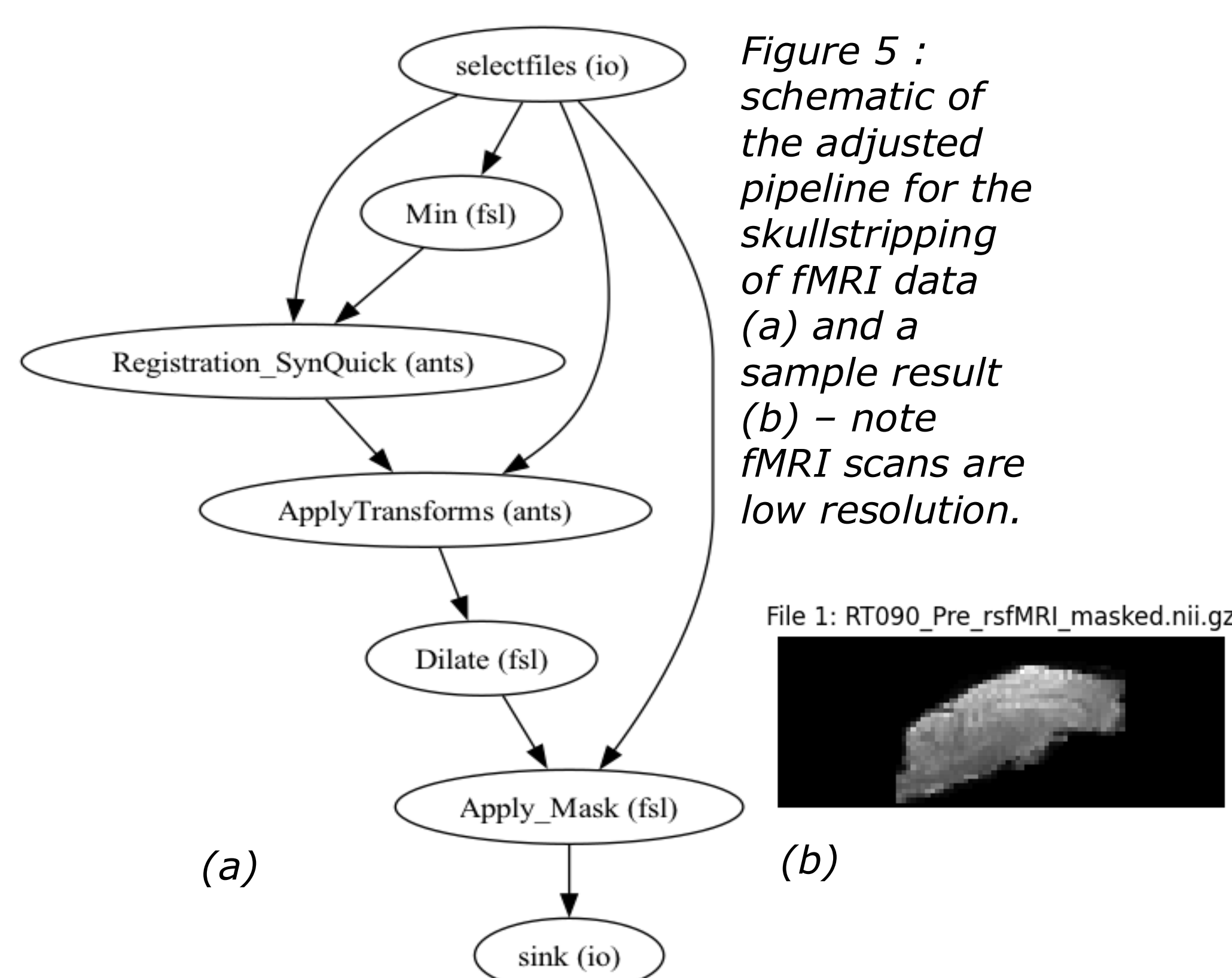


Figure 4 : comparison of skullstrip result from the pipeline (a) to FSL's BET tool (b)

References,
further information,
and example code
available via QR code to
GitHub repository :

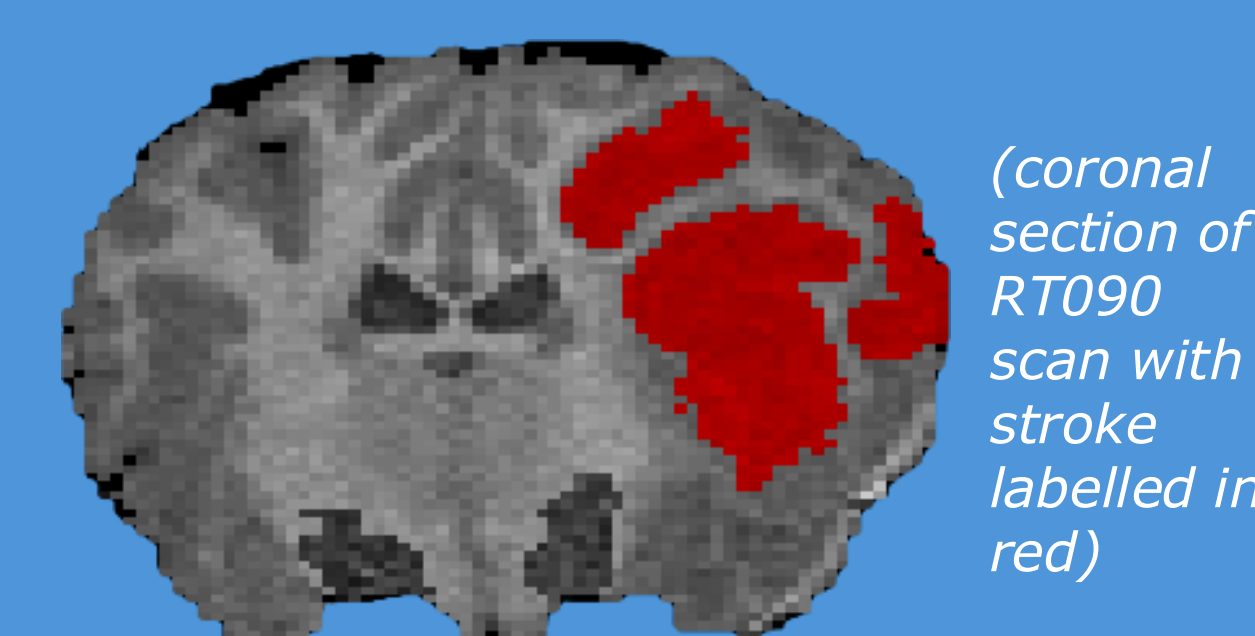


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Referencing done in Vancouver format



(coronal section of RT090 scan with stroke labelled in red)